

KAIST
2023 Fall
[MAS456] Statistical Methods with Computer
Homework 2

Munim Hasan Wasi

University: KAIST

Department: School of Computing

Student ID: 20210802

October 5, 2023

Contents

1	Chapter 4 - Classification	2
2	Chapter 5 - Resampling Methods	8

Chapter 4 - Classification

(1) #1

Given:

$$\text{Equation (4.2) : } p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$$

$$\text{Equation (4.3) : } \frac{p(X)}{1 - p(X)} = e^{\beta_0 + \beta_1 X}$$

Proof:

Starting with Equation (4.2):

$$p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$$

Now, let $F = \beta_0 + \beta_1 X$, so we have:

$$p(X) = \frac{e^F}{1 + e^F}$$

Now, to prove equivalence, we need to manipulate this expression to match Equation (4.3):

$$p(X)/(1 - p(X)) = e^F$$

First, let's find $1 - p(X)$:

$$\begin{aligned} 1 - p(X) &= \frac{1 + e^F - e^F}{1 + e^F} \\ 1 - p(X) &= \frac{1}{1 + e^F} \end{aligned}$$

Now, we can rewrite $p(X)/(1 - p(X))$:

$$\frac{p(X)}{1 - p(X)} = \frac{\frac{e^F}{1 + e^F}}{\frac{1}{1 + e^F}}$$

Now, simplify by multiplying both the numerator and denominator by $1 + e^F$:

$$\frac{e^F}{1} = e^F$$

So, we've shown that:

$$\frac{p(X)}{1 - p(X)} = e^F$$

Since we initially set $F = \beta_0 + \beta_1 X$, we can replace F with $\beta_0 + \beta_1 X$ to confirm the equivalence:

$$\frac{p(X)}{1 - p(X)} = e^{\beta_0 + \beta_1 X}$$

And that is Equation (4.3). Proven.

(2) #4

- (a) When we have a single feature X that is uniformly distributed on the interval $[0, 1]$, and we want to predict a test observation's response using observations within 10% of the range of X closest to the test observation, we are essentially looking at a small subinterval of width 0.1 centered around the test observation's X value. So, the average value for the whole range would be 10%. Some might think that the answer is 9.75% because the values in the range $[0, 0.05]$ cover less than 10% of the training data, but this is wrong. 10% is the correct answer.
- (b) In this case, we have two features X_1 and X_2 , both uniformly distributed on the interval $[0, 1]$. We want to predict a test observation's response using observations within 10% of the range of X_1 and 10% of the range of X_2 closest to the test observation. This creates a small square region in the (X_1, X_2) plane. The area of this square is $(0.1)^2 = 0.01$, which means we will, on average, use 1% of the available observations to make the prediction.
- (c) When we have 100 features, each is uniformly distributed on each feature, and again each feature ranges in value from 0 to 1. We wish to predict a test observation's response using observations within the 10% of each feature's range closest to that test observation. The volume of the hyperrectangle is $(0.1)^{100}$, which is an extremely small fraction of the total volume of the feature space. Consequently, we will use an extremely small fraction of the available observations to make the prediction which is $(0.1)^{98}\%$.
- (d) The drawback of KNN, when p is large, is that there are very few training observations "near" any given test observation. As p increases, the volume of the hyperrectangle that contains the nearby training observations becomes exponentially smaller relative to the total volume of the feature space. As we saw in parts (a)-(c), it becomes $(0.1)^{p-2}\%$ of the total space. This means that there are very few training observations that are close enough to provide meaningful information for prediction, leading to a deterioration in KNN's performance.
- (e) To create a p -dimensional hypercube centered around the test observation that contains, on average, 10% of the training observations, we need to find the side length of the hypercube for each value of p .
- For $p = 1$, the hypercube is a line segment. The length of each side of the hypercube is 0.1 since we want to capture 10% of the training observations along a single feature.
 - For $p = 2$, the hypercube is a square. The area of the square is 0.1, so each side has a length of $\sqrt{0.1} = 0.316$ since we want to capture 10% of the training observations in a two-dimensional space.
 - For $p = 100$, the hypercube is a 100-dimensional cube. The 'volume' of the hypercube is 0.1, so each side has a length of $(0.1)^{\frac{1}{100}}$ which is around 0.97.

In this case, for $p = 100$, the side length of the hypercube is 0.97 which is almost 1. So, to cover just 10% of training data, it basically needs to cover 97% of each direction. This illustrates the extreme sparsity problem that arises when dealing with high-dimensional feature spaces.

(3) #10 Using the fact that

$$\log \left(\frac{Pr(Y = k|X = x)}{Pr(Y = K|X = x)} \right) = \log \left(\frac{\pi_k f_k(x)}{\pi_K f_K(x)} \right)$$

and that everything can be considered as scalar for this exercise:

$$\begin{aligned} \log \left(\frac{Pr(Y = k|X = x)}{Pr(Y = K|X = x)} \right) &= \log \left(\frac{\pi_k \exp \left(-\frac{1}{2} \left(\frac{x - \mu_k}{\sigma} \right)^2 \right)}{\pi_K \exp \left(-\frac{1}{2} \left(\frac{x - \mu_K}{\sigma} \right)^2 \right)} \right) \\ &= \log \left(\frac{\pi_k}{\pi_K} \right) + \log \left(\frac{\exp \left(-\frac{1}{2} \left(\frac{x - \mu_k}{\sigma} \right)^2 \right)}{\exp \left(-\frac{1}{2} \left(\frac{x - \mu_K}{\sigma} \right)^2 \right)} \right) \\ &= \log \left(\frac{\pi_k}{\pi_K} \right) + \left(-\frac{1}{2} \left(\frac{x - \mu_k}{\sigma} \right)^2 \right) - \left(-\frac{1}{2} \left(\frac{x - \mu_K}{\sigma} \right)^2 \right) \\ &= \log \left(\frac{\pi_k}{\pi_K} \right) - \frac{1}{2} \left(\frac{x - \mu_k}{\sigma} \right)^2 + \frac{1}{2} \left(\frac{x - \mu_K}{\sigma} \right)^2 \\ &= \log \left(\frac{\pi_k}{\pi_K} \right) - \frac{1}{2\sigma^2} (x - \mu_k)^2 + \frac{1}{2\sigma^2} (x - \mu_K)^2 \\ &= \log \left(\frac{\pi_k}{\pi_K} \right) - \frac{1}{2\sigma^2} ((x - \mu_k)^2 - (x - \mu_K)^2) \\ &= \log \left(\frac{\pi_k}{\pi_K} \right) - \frac{1}{2\sigma^2} (x^2 - 2x\mu_k + \mu_k^2 - x^2 + 2x\mu_K - \mu_K^2) \\ &= \log \left(\frac{\pi_k}{\pi_K} \right) - \frac{1}{2\sigma^2} (2x(\mu_K - \mu_k) + \mu_k^2 - \mu_K^2) \\ &= \log \left(\frac{\pi_k}{\pi_K} \right) - \frac{\mu_k^2 - \mu_K^2}{2\sigma^2} + \frac{x(\mu_K - \mu_k)}{\sigma^2} \end{aligned}$$

From this line, we can interpret $a_k = \log \left(\frac{\pi_k}{\pi_K} \right) - \frac{\mu_k^2 - \mu_K^2}{2\sigma^2}$ and $b_k = \frac{\mu_K - \mu_k}{\sigma^2}$.

(4) #12

(a) The log odds of orange versus apple in my model is given by the logistic regression equation:

$$\log \left(\frac{Pr(Y = \text{orange}|X = x)}{Pr(Y = \text{apple}|X = x)} \right) = \log \left(\frac{\frac{\exp(\hat{\beta}_0 + \hat{\beta}_1 x)}{1 + \exp(\hat{\beta}_0 + \hat{\beta}_1 x)}}{\frac{1}{1 + \exp(\hat{\beta}_0 + \hat{\beta}_1 x)}} \right) = \log \left(\frac{\exp(\hat{\beta}_0 + \hat{\beta}_1 x)}{1} \right) = \hat{\beta}_0 + \hat{\beta}_1 x$$

So, the log odds of orange versus apple in my model is $\hat{\beta}_0 + \hat{\beta}_1 x$.

(b) The log odds of orange versus apple in my friend's model are given by the softmax formulation:

$$\log \left(\frac{Pr(Y = \text{orange}|X = x)}{Pr(Y = \text{apple}|X = x)} \right) = \hat{\alpha}_{\text{orange}0} - \hat{\alpha}_{\text{apple}0} + (\hat{\alpha}_{\text{orange}1} - \hat{\alpha}_{\text{apple}1})x$$

(c) In my model, if $\hat{\beta}_0 = 2$ and $\hat{\beta}_1 = -1$, the coefficients in my friend's model would be:

$$\hat{\alpha}_{\text{orange}0} - \hat{\alpha}_{\text{apple}0} = 2 \quad \text{and} \quad \hat{\alpha}_{\text{orange}1} - \hat{\alpha}_{\text{apple}1} = -1$$

But we can't solve for 4 unknowns from 2 equations.

(d) In my friend's model, if $\hat{\alpha}_{\text{orange}0} = 1.2$, $\hat{\alpha}_{\text{orange}1} = -2$, $\hat{\alpha}_{\text{apple}0} = 3$, and $\hat{\alpha}_{\text{apple}1} = 0.6$, the coefficients in my model would be:

$$\hat{\beta}_0 = \hat{\alpha}_{\text{orange}0} - \hat{\alpha}_{\text{apple}0} = 1.2 - 3 = -1.8$$

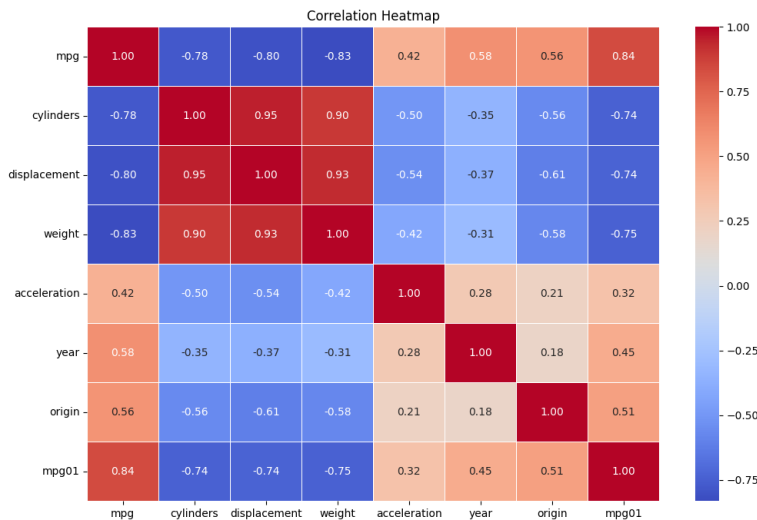
$$\hat{\beta}_1 = \hat{\alpha}_{\text{orange}1} - \hat{\alpha}_{\text{apple}1} = -2 - 0.6 = -2.6$$

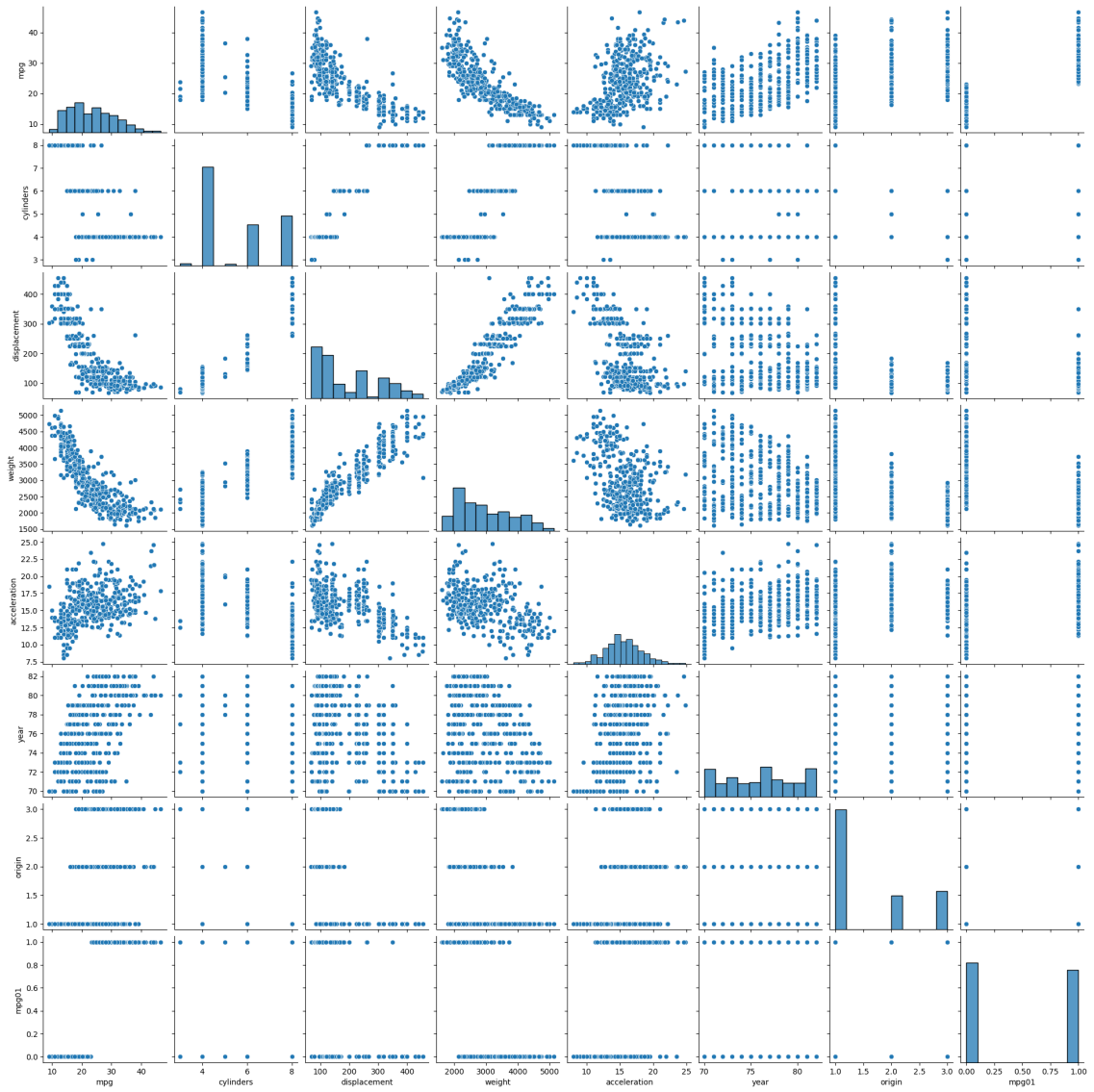
(e) As we can see from (a) - (d), the models are technically the same even though their parameters are stated differently. So their results should be the same as well. So the fraction should be 100%.

(5) #14

(a) Done

(b) From the scatterplots and correlation heatmaps, we can infer that the most correlated features that are likely to be useful in predicting mpg01 are cylinders, displacement and weight.





(c) Done

(d) Test error of LDA: 0.1742424242424242

(e) Test error of QDA: 0.13636363636363635

(f) Test error of Logistic Regression: 0.16666666666666663

(g) Test error of Naive Bayes: 0.1515151515151515

(h) K: 1

Error: 0.18181818181818177

K: 2

Error: 0.1742424242424242

K: 3

Error: 0.18939393939393945

K: 4

Error: 0.18939393939393945

K: 5

Error: 0.18181818181818177

K: 6
Error: 0.19696969696969702
K: 7
Error: 0.1742424242424242
K: 8
Error: 0.18181818181818177
K: 9
Error: 0.18939393939393945
K: 10
Error: 0.18939393939393945
Best K for KNN: 2
Test error of KNN with best K: 0.1742424242424242

Chapter 5 - Resampling Methods

- (6) #1 Using fundamental principles of variance and single-variable calculus, we aim to derive equation (5.6) and demonstrate that the value of α provided in (5.6) effectively minimizes the variance of $\alpha X + (1 - \alpha)Y$.

Equation 5.6 is expressed as:

$$\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}$$

Here are some relevant properties that I have used in this proof:

- The variance of a scaled random variable is given by $\text{Var}(aX) = a^2\text{Var}(X)$.
- The variance of the sum of two random variables is $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$.
- The covariance of scaled random variables is $\text{Cov}(aX, bY) = ab\text{Cov}(X, Y)$.

Now, σ_X^2 is the variance of X , σ_Y^2 is the variance of Y , and σ_{XY} is the covariance between X and Y . Then, we calculate the variance of $\alpha X + (1 - \alpha)Y$ as follows:

$$\text{Var}(\alpha X + (1 - \alpha)Y) = \alpha^2\sigma_X^2 + (1 - \alpha)^2\sigma_Y^2 + 2\alpha(1 - \alpha)\sigma_{XY}$$

This expression is then simplified to:

$$\alpha^2\sigma_X^2 + \sigma_Y^2 - 2\alpha\sigma_Y^2 + \alpha^2\sigma_Y^2 + 2\alpha\sigma_{XY} - 2\alpha^2\sigma_{XY}$$

To find the point where the rate of change of this function is zero with respect to α , we compute the partial derivative and set it to zero:

$$2\alpha\sigma_X^2 - 2\sigma_Y^2 + 2\alpha\sigma_Y^2 + 2\sigma_{XY} - 4\alpha\sigma_{XY} = 0$$

Rearranging the terms involving α leads us to the value of α that minimizes the variance:

$$\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}$$

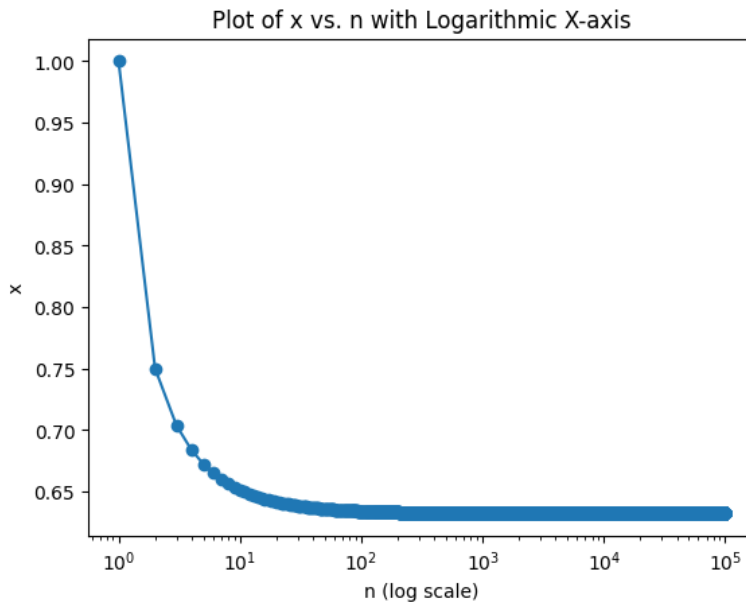
To confirm that this is indeed a minimum, we examine the second partial derivative with respect to α :

$$\frac{\partial^2}{\partial \alpha^2} = 2\sigma_X^2 + 2\sigma_Y^2 - 4\sigma_{XY} = 2(\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}) = 2\text{Var}(X - Y)$$

Since variance is always positive, this confirms that the second derivative is positive, indicating that the value of α given in (5.6) corresponds to a minimum.

- (7) #2

- (a) The probability that the first bootstrap observation is the j th observation from the original sample is $\frac{1}{n}$. So the probability that it is not the j th observation will be $1 - \frac{1}{n}$.
- (b) Same as in the previous case, which is $1 - \frac{1}{n}$. The reason is every bootstrap observation is a random sample, so the probability stays the same.
- (c) To find the probability that the j -th observation is absent from the bootstrap sample, we consider that it must not be selected for each of the n positions. Therefore, the probability is derived as $(1 - \frac{1}{n}) \cdot (1 - \frac{1}{n}) \cdot (1 - \frac{1}{n}) \dots n \text{ times} = (1 - \frac{1}{n})^n$.
- (d) When n equals 5, the likelihood that the j -th observation is present in the bootstrap sample is computed as $1 - (1 - \frac{1}{n})^n$, resulting in approximately 0.67232.
- (e) When n equals 100, the probability of the j -th observation being part of the bootstrap sample is $1 - (1 - \frac{1}{n})^n$, yielding approximately 0.6339677.
- (f) For n equals 10,000, the probability that the j -th observation is within the bootstrap sample is $1 - (1 - \frac{1}{n})^n$, resulting in approximately 0.6321224.
- (g) Graph:



The graph illustrates that as n increases, the probability converges rapidly toward 0.63.

- (h) In this problem, we are simulating the probability of having 4 when resampling numbers in the range $[0,100]$. The result of the simulation suggests a probability of approximately 0.6362. Which approximates the theoretical value of $1 - (1 - \frac{1}{100})^{100}$. It gives similar results for values other than 4 as well, as the selection of the number is arbitrary.

Note: There was an error in the code. I have used this corrected code for this simulation.

```
[10] rng = np.random.default_rng(10)
     store = np.empty(10000)
     for i in range(10000):
         draws = rng.choice(100, size=100, replace=True)
         store[i] = np.any(draws == 4)

     np.mean(store)

0.6362
```

(8) #5

(a) Done

(b) Validation Set Error: 0.035

(c) Validation Set Error: 0.035

Validation Set Error: 0.0344

Validation Set Error: 0.0342

As we can see, the error value changes on every run. This is because it depends on the split of the test and train data. As it is randomly chosen, each time the value is different.

(d) Validation Set Error: 0.0368

Validation Set Error: 0.0376

Validation Set Error: 0.0394

Including 'student' doesn't really change the error rate much. So we can say that there is almost no effect of including 'student' in the experiment.

(9) #6

(a) income 0.000005

balance 0.000227

(b) Done

(c) income 0.000004

balance 0.000211

(d) The values are approximately the same for both cases.

(10) #7

(a) Done

(b) Done

(c) Actual: Down

Prediction: Up

Is observation correctly classified? : No

(d) LOOCV Estimate for Test Error: 0.44995408631772266

Our error rate is slightly below 50% which is just a little bit better than a random coin toss.